Journal Pre-proof

Effects of pre-sleep protein consumption on muscle-related outcomes — A systematic review

Caio E.G. Reis, Laís M.R. Loureiro, Hamilton Roschel, Teresa H.M. da Costa

PII: S1440-2440(20)30719-2

DOI: https://doi.org/10.1016/j.jsams.2020.07.016

Reference: JSAMS 2368

To appear in: *Journal of Science and Medicine in Sport*

Received Date: 11 February 2020

Revised Date: 23 July 2020

Accepted Date: 28 July 2020

Please cite this article as: { doi: https://doi.org/

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2020 Published by Elsevier.

Journal Pre-proof

Review Article

Effects of pre-sleep protein consumption on muscle-related outcomes - a systematic review

Caio E. G. Reis[a]*

Laís M. R. Loureiro[b]

Hamilton Roschel[c]

Teresa H. M. da Costa[a]

[a] Nutritional Biochemistry Laboratory, Department of Nutrition, Universidade de Brasília, Brasília, DF, Brazil.

[b] Faculty of Health Sciences, Universidade de Brasilia, Brasília, DF, Brazil.

[c] Applied Physiology & Nutrition Research Group, School of Physical Education and Sport; Rheumatology Division; Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil.

* Corresponding author:

E-mail: caioedureis@gmail.com

Telephone: +55 61 981170044

Text-only word count: 2803.

Abstract word count: 256.

Number tables: 3.

Number of figures: 1.

Journal Pre-proof

Journal Pre-proof

Abstract

Objectives: The timing of protein intake over the day on muscle mass and strength gains have received interest in the literature. Thus, the aim of this systematic review is to analyze clinical studies that evaluated the acute effects of pre-sleep protein consumption on overnight muscle protein synthesis and the chronic effects on muscle mass and strength.

Method: A literature search was conducted up to June 2020 according to PRISMA statement and nine articles were included to analyze.

Results: The consumption of 20–40 g of casein approximately 30 minutes before sleep stimulates whole-body protein synthesis rates over a subsequent overnight period in young and elderly men (preceded or not by resistance exercise, respectively). In addition, pre-sleep protein consumption can augment the muscle adaptive response (muscle fiber cross-sectional area, strength and muscle mass) during 10–12 weeks of resistance exercise in young, but not in elderly men.

Conclusions: Based on current evidence, the consumption of 20–40 g of casein approximately 30 minutes before sleep improves protein synthetic response during an overnight recovery period in healthy young adult men, with possible positive effects on muscle mass and strength following prolonged resistance exercise. In elderly, despite the initial evidence regarding the pre-sleep protein enhances overnight muscle protein synthesis rates, the current available evidence is limited precluding to conclude about the chronic effects on skeletal muscle mass or strength. These conclusions need to be taken with caution due to uneven protein intakes between experimental groups. Therefore, more data are needed before further considering pre-sleep protein as an effective nutritional intervention.

Keywords: Sleep; Muscles; Exercise; Proteins; Sports medicine.

Journal Pre-proof

Journal Pre-proof

1. Introduction

Skeletal muscle mass is regulated by the daily fluctuation in rates of muscle protein synthesis (MPS) and breakdown (MPB). While the latter prevails in absence of nutrient intake, MPS is sensitive to both nutrition and exercise stimulus. In response to protein feeding, rates of MPS are significantly and transiently increased. Importantly, this effect is synergistically potentiated by resistance exercise (RE), further favoring a more positive protein balance and ultimately resulting in muscle mass accrual.[1,2]

Regarding dietary protein, the amount, quality, and timing of protein ingestion are major factors regulating muscle protein net balance. While the amount and type of protein necessary to maximally stimulate MPS are better understood in the literature,[3,4] questions around protein timing still remain. More recently, the "per meal" distribution of protein throughout the day has received increasingly interest in the literature. In accordance with this notion, adequate quantity and timing of protein ingestion could favor optimal response of the MPS machinery to repeated anabolic stimuli.[5,6] In fact, though not consensual,[7,8] there is evidence to suggest a significantly higher net positive protein balance in evenly distributed protein intake conditions when compared to protein-matched uneven distributions.[9-11]

Notably, net muscle protein balance remains negative in the absence of an adequate nutritional support, implying that the overnight period (usually the longest postabsorptive period of the day) may yield a negative net protein balance.[12] In this respect, pre-sleep protein intake has been considered an interesting approach to optimize protein balance, theoretically favoring muscle mass accretion.[13] Despite different studies have demonstrated that protein ingestion before sleep increases aminoacidemia during the night, stimulating MPS and suggesting a positive overnight net muscle protein balance;[14,15] there are some controversial results in the literature[16,17]. Thereby, up to date, there is no systematic review summarizing the effects of pre-sleep protein intake on muscle-related outcomes. Therefore, the aim of this review is to systematically assess the current literature and summarize the evidence regarding the effects of pre-sleep protein consumption on overnight muscle protein synthesis and its effects on muscle mass and strength in young adults and elderly.

Journal Pre-proof

Journal Pre-proof

2. Methods

2.1. Literature Searching

The initial search was conducted in December 2019 and updated in June 2020 to include all potential articles until this moment, according to the statement of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)[18] in the databases PubMed, Science Direct and Web of Science. The combinations of the following terms (in the title, abstract or in the keywords) were used: ('Sleep' OR 'Overnight') AND 'Muscle' AND 'Protein'. In order to reduce publication bias, the search was not restricted by language, publication date or any other filter. Additionally, we scrutinized references within identified papers. The inclusion criteria covered human clinical studies conducted with apparently healthy individuals (age > 19 years) that evaluated the acute effects of pre-sleep protein consumption on overnight MPS parameters or the chronic effects of pre-sleep protein intake (in association or not with exercise) on muscle mass and strength.

The literature search and study selection process were carried out by two independent investigators (C.E.G.R. and L.M.R.L.). We conducted an initial evaluation based on the title and abstract of the articles for assess if they meet the aims of the review (first screening). After, duplicate articles in more than one database were excluded. The methods section of each article was read (second screening) and those suitable for the review were assessed for eligibility by the full-text evaluation. The studies that did not match the inclusion criteria were excluded (with reasons). In cases where a consensus could not be reached, a third author (T.H.M.C.) made the decision regarding paper selection. Finally, the studies were included in qualitative synthesis.

2.2. Study selection and Data collection

Clinical trials were included if they met the following criteria: (1) sample of healthy adult/elderly individuals; (2) pre-sleep protein consumption versus a control treatment (placebo, carbohydrate or water); (3) assessed the effects of pre-sleep protein consumption on MPS parameters during overnight sleep and/or muscle mass and strength. These inclusion criteria were assessed by reading the respective study protocol. When necessary, additional data were requested from the study's corresponding author. The exclusion criteria were: (1) no pre-sleep protein consumption; (2)

Journal Pre-proof

Journal Pre-proof

any medication intake and/or diseases (3) no muscle parameters assessed; (4) reviews, abstracts, letters, conferences or seminar, personal opinions, thesis, and books.

The following information were recorded from the selected studies: authors, publication year, sample characteristics, study design, follow-up period, protein intake and exercise protocols, main results of MPS (whole-body protein synthesis – WBPS; and mixed-muscle protein fractional synthetic rate – FSR), and muscle mass and strength.

2.3. Summary measures

Data were summarized and presented in tables. Whole-body protein synthesis (WBPS) and mixed-muscle protein fractional synthetic rate (FSR) were reported in $\mu mol.kg^{-1}.h^{-1}$ and %/h, respectively. Studies reporting WBPS as $\mu mol.kg^{-1}.7.5h^{-1}$ or $\mu mol.kg^{-1}.min^{-1}$ were transformed to $\mu mol.kg^{-1}.h^{-1}$ by dividing its value per 7.5 or multiplying by 60, respectively. Data are presented as mean ± SEM (standard error of the mean) and $p \leq 0.05$ was considered to be significant.

3. Results

3.1. Literature search

The search strategy is summarized in **Figure 1**. The literature search identified 608 studies in the databases. After the first screening (based on title and abstract), 580 studies were removed. After removing duplicates (n = 4) and those not eligible due the inclusion criteria (n = 13) (second screening), we assessed the full-text for the eligible articles (n = 11) and ended up with nine trials. Two studies were excluded due to lack of control group (removed with reasons). Five out of the nine studies included assessed the acute effects of pre-sleep protein intake during overnight recovery sleep,[16,17,19–21] whereas four studies assessed the chronic effects (10-12 weeks of RE or endurance exercise training – ETT) on muscle mass and strength.[22-25] Three studies evaluated the acute effects of pre-sleep protein in healthy young adult men after an evening RE session,[16,17,19] whereas two studies assessed healthy elderly individuals with no RE.[20,21] Regarding the chronic studies, three were conducted with healthy young adult men (one in association with RE[22] and two combined with ETT,[23,24] and one with healthy elderly individuals (in association with RE).[25]

Journal Pre-proof

Figure 1.

3.2. Descriptive data and muscle-related outcomes

The relevant characteristics of the selected studies are summarized in **Table 1**. The acute response of WBPS and FSR extracted from the studies evaluating the effects of pre-sleep protein consumption on overnight MPS are presented in **Table 2**. All acute studies were performed using i.v. infusions with tracers of phenylalanine and tyrosine or leucine stable isotopes. The chronic response of pre-sleep protein consumption on muscle mass (quadriceps muscle cross-sectional area, type II fiber cross-sectional area, and whole-body lean mass) and strength gains (maximum dynamic strength) after 10-12 weeks of RE or ETT are presented in **Table 3**. For clarity purposes, results were sectioned according to sample characteristics (i.e.; young adults and elderly).

Table 1.

Table 2.

Table 3.

3.2.1. Young adult

Results are consistent in showing a significant increase on WBPS during post-exercise overnight recovery period after the consumption of 20–40 g of casein pre-sleep when RE was performed in the evening. Regarding FSR, only one[19] study showed a significant (though marginal) increase when compared with control group ($p = 0.05$).

Beelen et al. (2008) were the first to examine the impact of 0.3 $g \cdot kg^{-1}$ of casein (hydrolysate) supplementation (~21.4 g) on MPS rates during overnight recovery after a combined ETT and RE performed in the evening. WBPS rate over 9 hours recovery period was 19% higher in casein compared with control group ($p < 0.05$); however, FSR did not differ between groups ($p = 0.89$).[16] Res et al. (2012)[19] and Trommelen et al. (2018),[17] provided further evidence by assessing the effects of 40 and 30 g of casein, respectively, consumed 30 minutes before sleep on MPS during overnight

Journal Pre-proof

recovery (7.5 hours) after an evening RE session. Both studies showed a significant increase on WBPS rate in pre-sleep protein group compared with control ($p < 0.01$). Additionally, Trommelen et al. (2018) found no between-group difference on FSR ($p = 0.85$),[17] while Res et al. (2012) observed a significant increment of ~22% on FSR in the protein group *vs.* control ($p = 0.05$).[19]

Snijders et al.[22] evaluated the effects of 27.5 g of protein prior to sleep during a 12-wk resistance training program on muscle mass and strength. The authors found significant increases on muscle strength, quadriceps muscle cross-sectional area, and type II muscle fiber area in the pre-sleep protein group; however, daily protein intake was not matched across groups, meaning that the pre-sleep protein group had a significantly higher total protein intake compared with control (~1.9 ± 0.1 $g \cdot kg^{-1} \cdot day^{-1}$ *vs.* ~1.3 ± 0.1 $g \cdot kg^{-1} \cdot day^{-1}$, respectively; $p < 0.05$).

More recently, two studies evaluated the effects of pre-sleep casein consumption (~29 g) on skeletal muscle adaptations to ETT (10 – 12 weeks). Knuiman et al.[24] observed significantly greater whole body and leg lean mass accretion and whole body and trunk fat mass reduction in the protein group when compared with control (all $p < 0.05$) after 10 weeks of training. Conversely, Jonvik et al.[23] found no significant between-group differences in either whole body ($p = 0.43$) or leg lean mass ($p = 0.08$) following 12 weeks of training. It is important to note that both studies used the same pre-sleep protein dose, resulting in a significantly higher daily protein intake in the pre-sleep protein group compared with control (1.8 ± 0.1 $g \cdot kg^{-1} \cdot day^{-1}$ *vs.* 1.3 ± 0.1 $g \cdot kg^{-1} \cdot day^{-1}$ (21) and 1.58 ± 0.31 $g \cdot kg^{-1} \cdot day^{-1}$ *vs.* 1.19 ± 0.31 $g \cdot kg^{-1} \cdot day^{-1}$ (20); both $p < 0.01$).

3.2.2. Elderly

The studies with healthy elderly men assessed the acute effects of pre-sleep protein intake (without a RE stimulus) on MPS parameters during overnight sleep. The acute consumption of 40g of casein before sleep significantly increased WBPS and FSR rates compared with control group.[20,21] However, no chronic effects were observed on muscle mass and strength parameters.[25]

Groen et al. (2012) performed an intragastric protein administration during a 7-h overnight sleep in healthy elderly men to assess dietary protein digestion and absorption kinetics and subsequent MPS rates during overnight sleep. Forty grams of casein were administered via nasogastric stomach

Journal Pre-proof

feeding catheter 2 h after the subjects fell asleep. Muscle tissue biopsies were collected 3 h before and 5 h after the protein or control administration. Protein group showed increased overnight MPS rates (47.6% on WBPS, $p < 0.01$ and 55.2% on FSR rates, $p < 0.05$) compared with control.[20] This was the first study to show that dietary protein administration during sleep stimulates MPS in elderly people. In agreement, Kouw et al. (2017) showed that 40 g of casein consumed before sleep were able to increase MPS rates during overnight sleep by 18.9% on WBPS and 33.3% on FSR rates compared with control ($p < 0.05$) in healthy elderly men.[21] These findings extended the knowledge on pre-sleep (or during sleep) protein effects on MPS in healthy older men. However, whether the effectiveness of this strategy to positively modulate muscle mass and strength in elderly individuals remained to be answered.

Despite the encouraging results, a chronic study did not confirm this hypothesis. They showed that 21 g of leucine-enriched whey protein supplementation every night before sleep (and after each RE session) did not further enhance muscle mass and strength after 12 weeks of RE in healthy elderly, despite a higher protein consumption in protein *vs.* control (1.43 ± 0.04 $g.kg^{-1}.day^{-1}$ *vs.* 1.17 ± 0.06 $g.kg^{-1}.day^{-1}$, respectively; $p < 0.001$).[25]

4. Discussion

The present study systematically reviewed the current literature on the acute and chronic effects of pre-sleep protein consumption on muscle-related outcomes. In summary, the current data showed that the ingestion of 20 – 40 g of casein approximately 30 minutes before sleep stimulates WBPS rates over a subsequent overnight period (preceded or not by RE) in young and elderly individuals. Chronic pre-sleep protein consumption can also augment the muscle adaptive response (muscle fiber cross-sectional area, strength and muscle mass) during 12 weeks of RE in young, but not in elderly men. Moreover, pre-sleep protein in combination with ETT showed no additive effects on muscle in young individuals.

Dietary protein has been consistently demonstrated effective to induce significant and transient increases in MPS.[26] In this context, pre-sleep protein intake has been a promising strategy to induce a positive overnight net muscle protein balance, with possible repercussions on chronic muscle

Journal Pre-proof

adaptations. In fact, pre-sleep protein ingestion was shown to be effectively digested and absorbed, thereby increasing overnight plasma amino acid availability and stimulating MPS rates.[27] Theoretically, these results may contribute to a more practical advice, as pre-sleep strategy emerges as a potential nutrient timing strategy to maximize skeletal muscle hypertrophy.

Interestingly, despite the high dose of protein (casein or whey) consumed immediately before bed, gastroesophageal reflux was not reported in the studies included in this review. Although the short-meal-to-sleep interval favors reflux episodes, they are more prevalent in overweight individuals and related to fat intake.[28]

Notwithstanding the consistent increases observed for WBPS (and somewhat consistent for FSR), these results need to be taken with caution, as they do not necessarily translate to muscle mass accretion. WBPS and FSR rates are measured by the incorporation of stable isotope tracers of amino acids, and despite providing important and necessary mechanistic understanding regarding the acute protein synthetic response to nutrition and/or exercise, its findings may not be translated into a chronic setting. The use of labelled amino acids requires a well-controlled laboratory setting, allowing MPS to be assessed only over a few hours in a very restrictive environment, whereas the cumulative response to RE with adequate nutrition takes place following weeks of intervention in a free-living condition.[29–31] Indeed, Mitchell et al. (2014) showed no association between MPS (1–6 h) and changes in skeletal muscle hypertrophy over 16 weeks of RE in untrained men.[32]

Results from acute studies are, however, very encouraging and warrant further scrutinization. In this respect, current data exploring more long term effects of before-sleep protein intake are both scant and equivocal. Snijders et al. (2015) were the first to investigate the chronic effects of pre-sleep protein on muscle strength and mass accrual in young healthy men, with significant greater gains in the protein compared to control group.[22] The same group, however, found contrasting results on lean mass accretion when combining pre-sleep protein with ETT.[23,24] In addition, no effects of pre-sleep protein were found on muscle mass and strength in resistance training elderly men.[25]

Notably, a shared limitation across the chronic studies lies on significant between-group differences in total protein intake throughout the intervention[22–25] (Table 1), hampering further conclusions on the effect of protein timing rather than total amount. Although it is plausible to argue

Journal Pre-proof

Journal Pre-proof

that this limitation is mitigated by the fact that total protein intake was ≥ ~1.2 $g.kg^{-1}.day^{-1}$ in the placebo groups in studies with young population,[22–24] a recent meta-regression has shown protein to favor muscle adaptations up to a daily intake of ~1.6 $g.kg^{-1}.day^{-1}$,[33] suggesting that between-group differences may be, at least partially, related to unmatched total protein intake across experimental conditions. In respect of elderly population, it is possible to speculate that, given the anabolic resistance usually presented by this population,[34] the supplemental protein may have been insufficient to induce any additional effects beyond that of training.[25] Nonetheless, one may consider pre-sleep protein supplementation as a viable strategy to increase total daily protein intake, which may ultimately contribute to training-induced muscle adaptations.

It is undeniable, nonetheless, that nutrient timing is a growing field of investigation, and it may have important implications to overall health. Buckner et al. (2017) analyzed data from the U.S. National Health and Nutritional Examination Survey (1999–2002) and found greater leg lean mass and knee extensor strength in those consuming higher amounts of protein (20–30 g) in the evening than those who consumed protein in the afternoon.[35] These findings suggest that protein timing may provide a basis for interventional strategies to attenuate muscle mass loss in more clinically compromised groups,[36,37] although proper clinical evidence is lacking.

The results of this study can contribute to guide proper clinical recommendations about the pre-sleep protein intake on muscle mass and strength. We strongly suggest that future studies should focus on addressing important limitations of current literature, such as lack of protein-matched conditions, more robust sample sizes, and better understanding of the effects (if any) of different protein sources (when combined or not with exercise – RE and ETT) on acute and chronic settings in different populations, with special interest in more vulnerable individuals.

5. Conclusions

Based on the current state of evidence, the consumption of 20 – 40 g of protein approximately 30 minutes before sleep improves protein synthetic response during an overnight recovery period from evening RE in healthy young adult men, with possible positive effects on muscle mass and strength following prolonged RE. In elderly, despite the initial evidence of pre-sleep protein

Journal Pre-proof

enhancing overnight MPS, the current available evidence is limited, precluding further conclusions about its chronic effects on skeletal muscle mass or strength. Regarding endurance training, more studies are needed to understand the effects of pre-sleep protein intake on muscle adaptive response. Importantly, these conclusions are constrained by considerable study limitations, with special attention to uneven protein intakes between experimental groups, precluding any conclusion on the sole effect of timing (pre-sleep) of protein ingestion (rather than total amount) on muscle adaptations. Therefore, we contend that more data are needed before further considering pre-sleep protein as an effective nutritional intervention to maximize exercise-induced gains or mitigate age-related loss in muscle mass.

Funding

This research did not receive any specific grant from funding agencies.

Conflicts of interest

The authors declare no conflict of interest and no external financial support.

Acknowledgement

There is no acknowledgement needed.

Journal Pre-proof

Journal Pre-proof

References

1. Atherton PJ, Smith K. Muscle protein synthesis in response to nutrition and exercise. J Physiol. 2012;590(5):1049-1057.
2. Churchward-Venne TA, Burd NA, Phillips SM. Nutritional regulation of muscle protein synthesis with resistance exercise: strategies to enhance anabolism. Nutr Metab (Lond). 2012;9(1):40.
3. Burd NA, McKenna CF, Salvador AF, et al. Dietary protein quantity, quality, and exercise are key to healthy living: a muscle-centric perspective across the lifespan. Front Nutr. 2019;6:83.
4. Trommelen J, Betz MW, van Loon LJC. The muscle protein synthetic response to meal ingestion following resistance-type exercise. Sports Med. 2019;49(2):185-197.
5. Loenneke JP, Loprinzi PD, Murphy CH, et al. Per meal dose and frequency of protein consumption is associated with lean mass and muscle performance. Clin Nutr. 2016;35(6):1506-1511.
6. Schoenfeld BJ, Aragon AA. How much protein can the body use in a single meal for muscle-building? Implications for daily protein distribution. J Int Soc Sports Nutr. 2018;15:10.
7. Kim IY, Schutzler S, Schrader A, et al. Quantity of dietary protein intake, but not pattern of intake, affects net protein balance primarily through differences in protein synthesis in older adults. Am J Physiol Endocrinol Metab. 2015;308(1):E21-8.
8. Kim IY, Schutzler S, Schrader AM, et al. Protein intake distribution pattern does not affect anabolic response, lean body mass, muscle strength or function over 8 weeks in older adults: A randomized-controlled trial. Clin Nutr. 2018;37(2):488-493.
9. Moore DR, Areta JL, Coffey VG, et al. Daytime pattern of post-exercise protein intake affects whole-body protein turnover in resistance-trained males. Nutr Metab (Lond). 2012;9:91.

Journal Pre-proof

Journal Pre-proof

10. Areta JL, Burke LM, Ross ML, et al. Timing and distribution of protein ingestion during prolonged recovery from resistance exercise alters myofibrillar protein synthesis. J Physiol. 2013;591(Pt 9):2319-2331.
11. Mamerow MM, Mettler JA, English KL, et al. Dietary protein distribution positively influences 24-h muscle protein synthesis in healthy adults. J Nutr. 2014;144(6):876-880.
12. Stokes T, Hector AJ, Morton RW, et al. Recent perspectives regarding the role of dietary protein for the promotion of muscle hypertrophy with resistance exercise training. Nutrients. 2018;10(2):E180.
13. Trommelen J, van Loon LJC. Pre-sleep protein ingestion to improve the skeletal muscle adaptive response to exercise training. Nutrients. 2016;8(12):E763.
14. Trommelen J, Holwerda AM, Kouw IW, et al. Resistance exercise augments postprandial overnight muscle protein synthesis rates. Med Sci Sports Exerc. 2016;48(12):2517-2525.
15. Holwerda AM, Kouw IW, Trommelen J, et al. Physical activity performed in the evening increases the overnight muscle protein synthetic response to presleep protein ingestion in older men. J Nutr. 2016;146(7):1307-14.
16. Beelen M, Tieland M, Gijsen AP, et al. Coingestion of carbohydrate and protein hydrolysate stimulates muscle protein synthesis during exercise in young men, with no further increase during subsequent overnight recovery. J Nutr. 2008;138(11):2198-204.
17. Trommelen J, Kouw IWK, Holwerda AM, et al. Pre-sleep dietary protein-derived amino acids are incorporated in myofibrillar protein during post-exercise overnight recovery. Am J Physiol Endocrinol Metab. 2018;314(5):E457-E467.
18. Moher D, Liberati A, Tetzlaff J, et al; PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. J Clin Epidemiol. 2009;62(10):1006-12.
19. Res PT, Groen B, Pennings B, et al. Protein ingestion before sleep improves postexercise overnight recovery. Med Sci Sports Exerc. 2012;44(8):1560-9.

Journal Pre-proof

Journal Pre-proof

20. Groen BB, Res PT, Pennings B, et al. Intragastric protein administration stimulates overnight muscle protein synthesis in elderly men. Am J Physiol Endocrinol Metab. 2012;302(1):E52-60.
21. Kouw IW, Holwerda AM, Trommelen J, et al. Protein ingestion before sleep increases overnight muscle protein synthesis rates in healthy older men: a randomized controlled trial. J Nutr. 2017;147(12):2252-2261.
22. Snijders T, Res PT, Smeets JS, et al. Protein ingestion before sleep increases muscle mass and strength gains during prolonged resistance-type exercise training in healthy young men. J Nutr. 2015;145(6):1178-84.
23. Jonvik KL, Paulussen KJM, Danen SL, et al. Protein supplementation does not augment adaptations to endurance exercise training. Med Sci Sports Exerc. 2019;51(10):2041-2049.
24. Knuiman P, van Loon LJC, Wouters J, et al. Protein supplementation elicits greater gains in maximal oxygen uptake capacity and stimulates lean mass accretion during prolonged endurance training: a double-blind randomized controlled trial. Am J Clin Nutr. 2019;110(2):508-518.
25. Holwerda AM, Overkamp M, Paulussen KJM, et al. Protein supplementation after exercise and before sleep does not further augment muscle mass and strength gains during resistance exercise training in active older men. J Nutr. 2018;148(11):1723-1732.
26. Gorissen SH, Rémond D, van Loon LJ. The muscle protein synthetic response to food ingestion. Meat Sci. 2015;109:96-100.
27. Snijders T, Trommelen J, Kouw IWK, et al. The impact of pre-sleep protein ingestion on the skeletal muscle adaptive response to exercise in humans: an update. Front Nutr. 2019 Mar 6;6:17.
28. Surdea-Blaga T, Negrutiu DE, Palage M, Dumitrascu DL. Food and gastroesophageal reflux disease. Curr Med Chem. 2019; 26(19): 3497-3511.
29. Wilkinson DJ, Brook MS, Smith K, et al. Stable isotope tracers and exercise physiology: past, present and future. J Physiol. 2017;595:2873-2882.

Journal Pre-proof

Journal Pre-proof

30. Smeuninx B, McKendry J. Mechanisms of resistance exercise-induced muscle hypertrophy: 'You can't make an omelette without breaking eggs'. J Physiol. 2016;594:7159-7160.
31. Wilkinson DJ, Franchi MV, Brook MS, et al. A validation of the application of D(2)O stable isotope tracer techniques for monitoring day-to-day changes in muscle protein subfraction synthesis in humans. Am J Physiol Endocrinol Metab. 2014;306(5):E571-579.
32. Mitchell CJ, Churchward-Venne TA, Parise G, et al. Acute post-exercise myofibrillar protein synthesis is not correlated with resistance training-induced muscle hypertrophy in young men. PLoS One. 2014;9(2):e89431.
33. Morton RW, Murphy KT, McKellar SR, et al. A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. Br J Sports Med. 2018;52(6):376-84.
34. Burd NA, Gorissen SH, van Loon LJ. Anabolic resistance of muscle protein synthesis with aging. Exerc Sport Sci Rev. 2013 Jul;41(3):169-73.
35. Buckner SL, Loenneke JP, Loprinzi PD. Protein timing during the day and its relevance for muscle strength and lean mass. Clin Physiol Funct Imaging. 2018;38(2):332-7.
36. Murphy CH, Oikawa SY, Phillips SM. Dietary protein to maintain muscle mass in aging: a case for per-meal protein recommendations. J Frailty Aging. 2016;5(1):49-58.
37. Morton RW, Traylor DA, Weijs PJM, et al. Defining anabolic resistance: implications for delivery of clinical care nutrition. Curr Opin Crit Care. 2018;24(2):124-130.

Journal Pre-proof

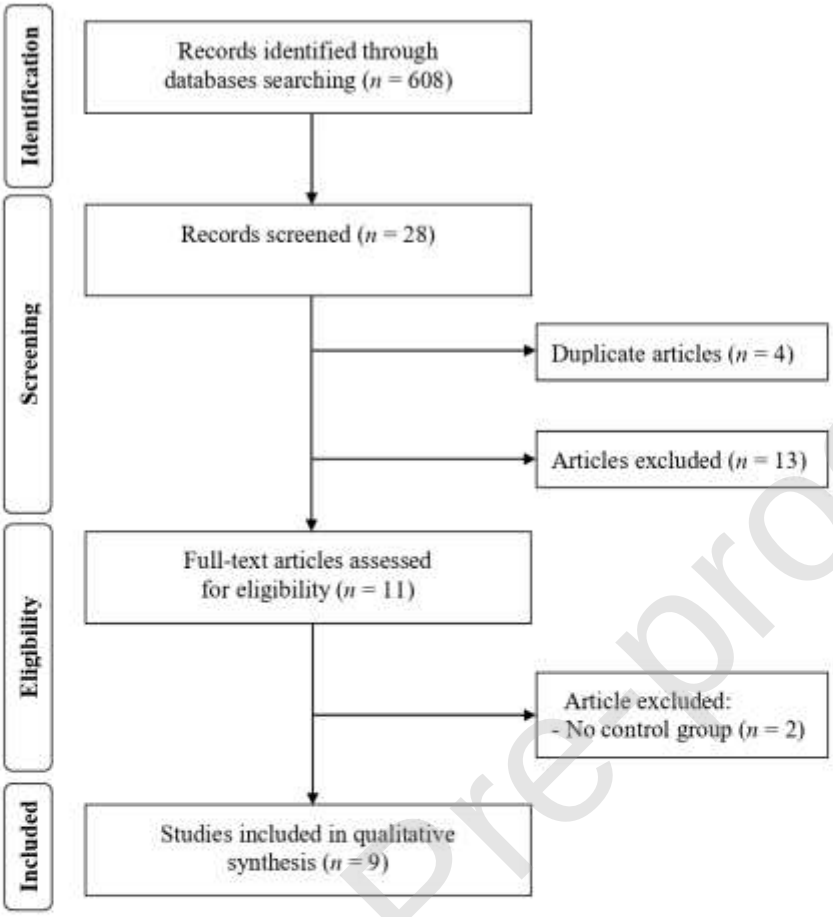
Journal Pre-proof

Figure legend

Figure 1. Study-selection scheme identifying clinical trials that evaluated the effects of pre-sleep protein consumption on overnight muscle protein synthesis or the chronic effects of pre-sleep protein intake on muscle mass and strength.

Journal Pre-proof

Journal Pre-proof



Journal Pre-proof

Journal Pre-proof

Table 1. Descriptive characteristics of the selected studies on acute and chronic effects of pre-sleep protein consumption on muscle-related outcomes

Author, year	**Sample size (mean age)**	**Study Design**	**Standardized diet ([a]protein g.kg^{-1}.d^{-1})**	**Exercise Protocol (min)**	**Protein source**	**Protein* (g; g.kg^{-1}.d^{-1})**	**Period**
Acute response							
Young adult							
Beelen et al., 2008[16]	20 (21)[¥]	DBRCT	Yes (1.1)	40'ETT+80'RE	Casein hydrolysate and carbohydrate	21.4g; 0.30†	9.0h
Res et al., 2012[19]	15 (22)[¥]	RCT	Yes (1.2)	45'RE	Casein	40.0g; 0.54	7.5h
Trommelen et al., 2018[17]	24 (23)[¥]	DBRCT	Yes (1.2)	60'RE	Casein	30.0g; 0.42	7.5h
Elderly							
Groen et al., 2012[20]	16 (74)	RCT	Yes (0.9)	No	Casein	40.0g; 0.50	5.0h
Kouw et al., 2017[21]	48 (72)	DBRCT	Yes (1.1)	No	Casein	40.0g; 0.50	7.5h
Chronic response							
Young adult							
Snijders et al., 2015[22]	39 (22)[¥]	DBRCT	No (1.9)[€]	60'RE (3x/wk)	Casein and casein hydrolysate	27.5g; 0.36	12wks

Journal Pre-proof

Jonvik et al., 2019[23]	60 (27)[¥]	DBRCT	No (1.6)[€]	60'ETT (3x/wk)	Casein	28.7g; 0.36	12wks
Knuiman et al., 2019[24]	40 (22)[¥]	DBRCT	No (1.8)[€]	60'ETT (3x/wk)	Casein	28.7g; 0.38	10wks
Elderly							
Holwerda et al., 2018[25]	41 (70)	DBRCT	No (1.4)[€]	60'RE (3x/wk)	Leucine-enriched whey protein	20.7g; 0.27	12wks

[¥]: Recreationally active volunteers; DBRCT: Double-blind randomized clinical trial; RCT: Randomized clinical trial; [a]: Protein intake ($g.kg^{-1}.d^{-1}$) in protein group; [€]: Protein intake including supplement were higher in protein group compared with control ($p < 0.05$); ETT: Endurance exercise training; RE: Resistance exercise training; *: Diet with an additional pre-sleep protein intake; †: Data provided by the authors.

Journal Pre-proof

Journal Pre-proof

Table 2. Acute effects of pre-sleep protein consumption on muscle protein synthesis parameters during overnight period

Author, year	**WBPS (µmol.kg^{-1}.h^{-1})**		**FSR (%/h)**	
	Protein	**Control**	**Protein**	**Control**
Young adult				
Beelen et al., 2008[16]	34.2 ± 1.4*	28.6 ± 1.3	0.056 ± 0.004	0.057 ± 0.004
Res et al., 2012[19]	41.5 ±1.0**	32.8 ± 1.2	0.059 ± 0.005*	0.048 ± 0.004
Trommelen et al., 2018[17]	53.0 ± 2.0**	-6.0 ± 1.0	0.055 ± 0.002	0.057 ± 0.002
Elderly				
Groen et al., 2012† [20]	42.8 ± 1.9**	29.0 ± 0.6	0.045 ± 0.002*	0.029 ± 0.002
Kouw et al., 2017† [21]	37.2 ± 1.2*	31.3 ± 0.8	0.044 ± 0.003*	0.033 ± 0.002

Data are presented as mean ± SEM; WBPS: Whole-body protein synthesis; FSR: Mixed-muscle protein fractional synthetic rate; †: Data provided by the authors; *: $p \leq 0.05$ when compared with control; **: $p < 0.01$ when compared with control.

Journal Pre-proof

Journal Pre-proof

Table 3. Chronic effects of pre-sleep protein consumption on muscle mass and strength after 10-12 weeks of exercise training

Author, year	**Δ Leg CSA (cm^2)**		**Δ Type II FCSA (μm^2)**		**Δ Muscle strength (kg)**		**Δ WB lean mass (kg)**	
	Protein	**Control**	**Protein**	**Control**	**Protein**	**Control**	**Protein**	**Control**
Young adult								
Snijders et al., 2015[22]	8.4 ± 1.1*	4.8 ± 0.8	2319 ± 368*	1017 ± 353	164 ± 11**	130 ± 9	NA	NA
Jonvik et al., 2019[23]	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	0.5 ± 1.1	-0.1 ± 1.1
Knuiman et al., 2019[24]	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	1.5** ± N.I.	0.4 ± N.I.
Elderly								
Holwerda et al., 2018† [25]	3.9 ± 0.5	5.6 ± 0.8	891 ± 332	1005 ± 443	17 ± 3	16 ± 3	NA	NA

Data are presented as mean ± SEM; Leg CSA: Quadriceps muscle cross-sectional area; FCSA: Vastus lateralis fiber cross-sectional area; WB: Whole-body;

†: data provided by the authors; N.A.: Not applicable - data not evaluated; *: $p < 0.05$ when compared with control. **: $p < 0.01$ when compared with control.

Journal Pre-proof